

WEEK 8 – POWER BI STUDENT PERFORMANCE DASHBOARD

1. Introduction

In Week 8, an interactive **Student Performance Dashboard** was created using Microsoft Power BI. The dashboard was developed to analyze students' performance in **Math, Reading, and Writing** and to compare their scores across different categories. The dashboard presents the analysis through **KPI cards, charts, slicers, and interactive visualizations**, making it easier to understand overall student performance and identify important patterns.

2. Dashboard Objectives

The main objectives of the dashboard are:

- To display the overall average Math, Reading, and Writing scores
- To calculate and display the overall average score.
- To analyze performance based on race/ethnicity.
- To compare scores based on parental education level.
- To analyze performance according to lunch category.
- To compare students based on test preparation.
- To analyze performance based on gender.
- To present the results using interactive Power BI visuals.

3. KPI Cards

Five KPI cards were created at the top of the dashboard to provide a quick summary of student performance.

The dashboard displays:

Average Math Score – 66.09
Average Reading Score – 69.17
Average Writing Score – 68.05
Average Score – 67.77
Overall Pass % – 0.97

These KPIs provide an immediate overview of the students' overall academic performance.

Average Math Score
66.09

Average Reading Score
69.17

Average Writing Score
68.05

Average Score
67.77

Overall Pass %
0.97

4. Performance by Race/Ethnicity

A **clustered column chart** was created to compare the following scores across different race/ethnicity groups:

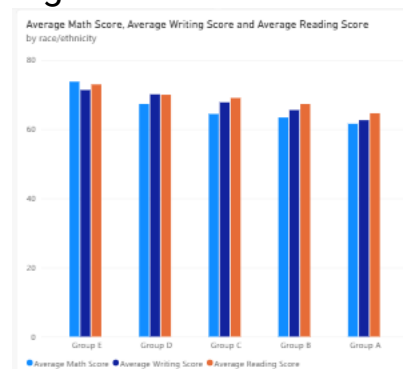
- Average Math Score

- Average Reading Score

- Average Writing Score

The groups included are **Group A, Group B, Group C, Group D, and Group E.**

The visualization helps compare academic performance between the different groups and identify which groups have relatively higher or lower scores.



5. Average Score by Parental Level of Education

A **donut chart** was used to display the Average Score according to the **parental level of education**.

The categories represented in the dashboard include:

- Master's Degree

- Bachelor's Degree

- Associate's Degree

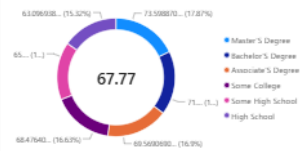
- Some College

- Some High School

- High School

This visualization helps examine the distribution of student performance across different parental education levels.

Average Score
by parental level of education



6. Performance by Lunch

A **treemap** was created to analyze student performance based on lunch category.

The categories are:

Standard

Free/Reduced

The dashboard displays:

Average Reading Score

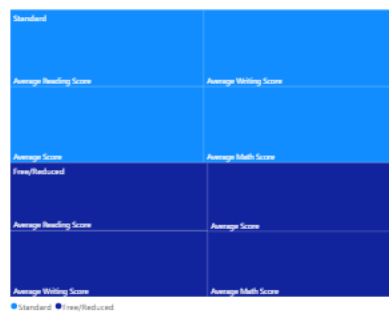
Average Writing Score

Average Score

Average Math Score

The visualization allows the performance of the two lunch categories to be compared easily.

Average Math Score, Average Reading Score, Average Score and Average Writing Score
by lunch



7. Performance by Test Preparation Course

An **area chart** was used to compare students based on whether they completed the test preparation course.

The two categories are:

Completed

None

The chart displays:

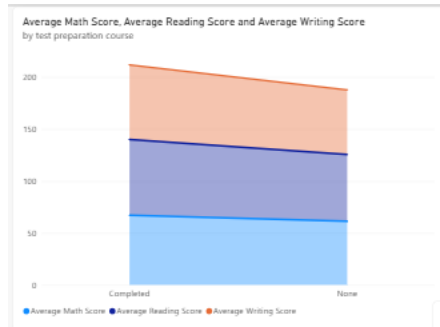
Average Math Score

Average Reading Score

Average Writing Score

This visualization helps identify differences in academic

performance between students who completed test preparation and those who did not.



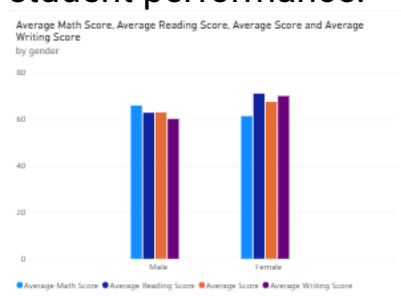
8. Performance by Gender

A **clustered column chart** was created to compare the performance of **Male and Female** students.

The following measures were included:

- Average Math Score
- Average Reading Score
- Average Score
- Average Writing Score

The chart provides a subject-wise comparison of male and female student performance.



9. Overall Performance by Lunch

A **stacked column chart** was used to compare Standard and Free/Reduced lunch categories.

The chart includes:

- Average Score
- Average Reading Score
- Average Writing Score
- Average Math Score

This provides an overall visual comparison of academic performance between the two lunch categories.

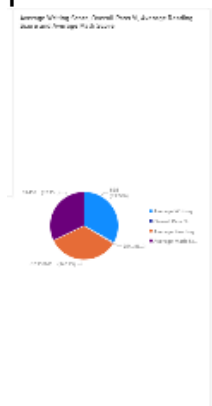


10. Overall Score Distribution

A **pie chart** was created to display the distribution of the main performance measures:

Average Writing Score
 Overall Pass %
 Average Reading Score
 Average Math Score

This gives a summarized visual representation of the key performance indicators used in the dashboard.



11. Slicers

Interactive filtering was added to the dashboard to allow users to analyze specific groups.

The dashboard includes filters for:

Race/Ethnicity
 Lunch
 Parental Level of Education

These filters make the dashboard interactive and allow users to

view the performance of selected categories.



12. Key Insights

The dashboard provides the following major insights:

The **overall average score is 67.77** .

The **average Reading Score is the highest at 69.17** .

The **average Math Score is 66.09** .

The **average Writing Score is 68.05** .

Student performance varies across different race/ethnicity groups.

Performance can be compared across different parental education levels.

The Standard and Free/Reduced lunch categories show differences in student performance.

Students who completed the test preparation course can be compared with students who did not.

Gender-wise performance varies across Math, Reading, and Writing.

Interactive slicers make it easier to analyze specific student groups.

13. Conclusion

The Week 8 Power BI dashboard provides an **interactive and visual analysis of student performance** . Different charts and KPI cards were used to compare Math, Reading, and Writing scores across demographic and educational categories.

The dashboard makes it easier to identify **performance differences, compare student groups, and understand overall academic trends** .